MATHEMATICAL MODELS & METRICS SPECIFICATIONS

TrustInterview AI Math Engine

An Analysis of Cumulative, Aggregated, and Normalized Scoring Systems

1. Cumulative Running Average Model

To optimize database read/write cycles, the system computes the candidate's historical average score incrementally upon completing each new session. This eliminates the need to load and sum every past session score array on simple score submissions.

Formula: $A_n = [\,A_{(n-1)} * (n - 1) + S_n\,] / n$

Variable Definitions

- **An:** The updated cumulative average score recorded in the database after the current session.
- **A(n-1):** The previously saved running average score (re-used as the baseline aggregate).
- **n:** The total count of graded sessions completed by the candidate including the current run.
- **Sn:** The raw, out-of-10 average score evaluated by the LLM for the current interview session.

Algorithm & Code Implementation

During the score submission callback, the previous average is retrieved and factored into the new session count weight. This provides O(1) time complexity calculations.

```
// question-ledger.js:recordScore
ledger[key].totalSessions = (ledger[key].totalSessions || 0) + 1;
const sessions = ledger[key].totalSessions;
const prev = ledger[key].avgScore || 0;
ledger[key].avgScore = parseFloat(((prev * (sessions - 1) + score) /
sessions).toFixed(2));
```

2. Segregated Data Aggregation Model

For multi-tenant recruiter isolation (e.g. Google vs. Microsoft dashboards), the system moves from unified averages to segregated sub-aggregates. The backend filters the session array by the active tenant ID and applies standard statistical models.

Arithmetic Mean: $Average = (\text{Sum of } S_i \text{ for } i = 1 \text{ to } k) / k$

Maximum Value: $Best\ Score = Max(S_1, S_2, ..., S_k)$

Variable Definitions

- **k:** The total number of interview sessions completed specifically for the queried company.
- **Si:** The score of the i-th session belonging to the target company filter.

Usage & Security Isolation

When a Google Recruiter logs in, only Google sessions (k) are loaded. Candidate scores are isolated, preventing other company campaign results from leaking onto the current dashboard.

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3. Linear Normalization Model

Raw interview scores are outputted by the grading agents out of 10. To maintain a standardized, recruiter-friendly scoring experience, these values undergo a linear percentage mapping on frontend dashboards.

Formula: $$\text{Percentage} = (\text{Score} / 10) * 100\% = \text{Score} * 10\%$$

Usage

Normalizes raw values like 8.2 to 82% dynamically. This maintains grade visibility across proctoring, average stats cards, and selection badges.